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CUET UG

Physics (322)

Electromagnetic Induction and Alternating Currents

Official Syllabus Unit 4 · Class 12 NCERT · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA Physics-322, Unit 4)	Key Exam Focus & Skills
Electromagnetic Induction & Faraday's Laws	Electromagnetic induction; Faraday's laws, induced emf and current; Lenz's Law, Self and mutual induction.	Flux $\Phi = BA\cos\theta$, Faraday's law $e = -N\left(d\frac{\Phi}{d}t\right)$, induced charge $q = \Delta\frac{\Phi}{R}$ independent of Δt, Lenz's law and energy conservation.
Motional EMF & Eddy Currents	Faraday's laws, induced emf and current; Lenz's Law.	Translational motional EMF $e = Bvl$, rotating rod $e = \frac{1}{2}B\omega l^2$, eddy current damping, laminated core reduction.
Self & Mutual Inductance	Self and mutual induction.	$L = \mu_0 n^2 Al$, $M = \mu_0 n_1 n_2 Al$, coupling $k = \frac{M}{\sqrt{L_1 L_2}}$, magnetic energy $U = \frac{1}{2}LI^2$, energy density $u_B = \frac{B^2}{2\mu_0}$.
AC Generation & RMS Fundamentals	Alternating currents, peak and rms value of alternating current/voltage; AC generator.	Generator peak EMF $e_0 = NBA\omega$, sinusoidal wave $e = e_0\sin(\omega t)$, $V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$, $I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$, half-cycle mean $2\frac{I_0}{\pi}$.
Reactance, Impedance & LCR Circuits	Reactance and impedance; LCR series circuit, phasors, resonance.	$X_L = \omega L$, $X_C = \frac{1}{\omega C}$, impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$, resonance $\omega_r = \frac{1}{\sqrt{LC}}$, $Q = \left(\frac{1}{R}\right)\sqrt{\frac{L}{C}} = \frac{\omega_r}{\Delta\omega}$.
Power in AC & Transformers	Power in AC circuits, power factor, wattless current; Transformer.	$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}}\cos\varphi$, power factor $\cos\varphi = \frac{R}{Z}$, wattless current $I_{\text{rms}}\sin\varphi$, transformer ratio $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} = k$, eddy/copper losses.

EXAM PATTERN

CUET UG Exam Architecture: Unit 4 accounts for roughly 10% to 12% of the CUET UG Physics test (5 to 6 MCQs). Standard question archetypes include motional EMF across rotating conductors, resonant frequency / quality factor scaling, wattless current condition, and step-up/down transformer turns-ratio calculations.

Electromagnetic Induction & Faraday's Laws

Magnetic Flux (Φ_B)

Magnetic flux measures the total number of magnetic field lines passing through a given surface area (A):

MAGNETIC FLUX DEFINITION

$$\Phi_B = \int (B) \cdot d(A) = BA\cos\theta \quad [\text{SI Unit: Weber (Wb) or T} \cdot \text{m}^2]$$

where θ is the angle between the magnetic field vector (B) and the outward area normal vector $\hat{n}$.

- **Dimensional formula:** $[\Phi_B] = [ML^2T^{-2}A^{-1}]$.
- When (B)$\perp$ surface ($\theta = 0°$), $\Phi_B = BA$ (maximum).
- When (B) $\parallel$ surface ($\theta = 90°$), $\Phi_B = 0$.

Faraday's Laws of Electromagnetic Induction

- **First Law (Qualitative):** Whenever the magnetic flux linked with a closed electrical circuit changes with time, an electromotive force (EMF) is induced in the circuit. The induced EMF persists as long as the flux variation continues.
- **Second Law (Quantitative):** The magnitude of the induced EMF is directly proportional to the time rate of change of magnetic flux linked with the circuit.

FARADAY-LENZ LAW OF INDUCTION

For a closely wound coil of N turns:

$$e = -N\frac{d\Phi_B}{dt}$$

If the total resistance of the closed loop is R:

$$I = \frac{e}{R} = -\frac{N}{R}\frac{d\Phi_B}{dt}$$

Induced Electric Charge (Δq):

$$\Delta q = \int I dt = \frac{1}{R}\int |d\Phi_B| = \frac{\Delta\Phi_B}{R}$$

Crucial CUET Result: The total induced charge flowing through a circuit depends **only** on the total change in flux ($\Delta\Phi_B$) and loop resistance (R), and is **completely independent** of the time taken (Δt) for the change to occur.

WORKED EXAMPLE 1: INDUCED EMF AND CHARGE IN COLLAPSING FIELD

Problem: A circular coil of radius $r = 10$ cm with $N = 500$ turns and total resistance $R = 20\Omega$ is placed perpendicular to a uniform magnetic field $B = 0.40$ T. If the field drops

uniformly to zero in $\Delta t = 0.050$ s, determine: (a) the magnitude of induced EMF, (b) the induced current, and (c) the total charge circulated.

Solution: Area of circular coil: $A = \pi r^2 = \pi(0.10)^2 = 0.01\pi$ m$^2 \approx 3.14 \times 10^{-2}$ m^2. Initial flux per turn: $\Phi_1 = BA = 0.40 \times 3.14 \times 10^{-2} = 1.257 \times 10^{-2}$ Wb. Final flux: $\Phi_2 = 0$. Change in flux per turn: $|\Delta\Phi| = 1.257 \times 10^{-2}$ Wb. (a) Induced EMF:

$$|e| = N\frac{|\Delta\Phi|}{\Delta t} = 500 \times \frac{1.257 \times 10^{-2}}{0.050} = 125.7 \text{ V}$$

(b) Induced current:

$$I = |e\frac{|}{R} = \frac{125.7}{20} = 6.285 \text{ A}$$

(c) Total charge passed:

$$\Delta q = \frac{N\ |\Delta\Phi|}{R} = \frac{500 \times 1.257 \times 10^{-2}}{20} = 0.314 \text{ C}$$

(Verification: $\Delta q = I\Delta t = 6.285 \times 0.050 = 0.314$ C).

Lenz's Law, Motional EMF & Eddy Currents

Lenz's Law & Conservation of Energy

Lenz's Law gives the direction of the induced EMF and current: > **Statement:** The polarity of induced EMF is always such that it tends to produce an induced current whose magnetic field opposes the change in magnetic flux that produces it.

The negative sign in $e = -(d\frac{\Phi}{d}t)$ mathematically represents Lenz's Law.

- **Energy Conservation Justification:** When a magnet is pushed toward a closed loop, the induced current creates a repulsive magnetic pole. External mechanical work must be done against this repulsive force to move the magnet. This mechanical work is converted precisely into electrical and thermal Joule energy in the loop.
- If induced current aided the flux change, a perpetual motion machine of the first kind would arise, violating the First Law of Thermodynamics.

Motional Electromotive Force

When a conductor moves across a magnetic field, the magnetic Lorentz force on free conduction electrons $((F)_m = -e((v) \times (B)))$ separates charges, establishing an induced EMF.

1. Translational Conductor: Straight rod of length l moving with velocity (v) perpendicular to a uniform field (B):

$$e = (v) \times (B) \cdot (l) = Bvl$$

Induced current: $I = \frac{Bvl}{R}$. External mechanical force required:

$$F = IlB = \frac{B^2l^2v}{R}$$

2. Rotating Conductor: Rod of length l rotated with angular speed ω about one end perpendicular to field (B): Linear velocity at distance x: $v(x) = \omega x$.

$$e = \int_0^l Bv(x)dx = B\omega\int_0^l xdx$$

$$e = \frac{1}{2}B\omega l^2 = \frac{1}{2}B(2\pi f)l^2 = B\pi fl^2$$

Mechanical power input:

$$P = Fv = \frac{B^2 l^2 v^2}{R} = I^2 R$$

Since $\pi l^2 = A$ (area swept per revolution):

$$e = BfA$$

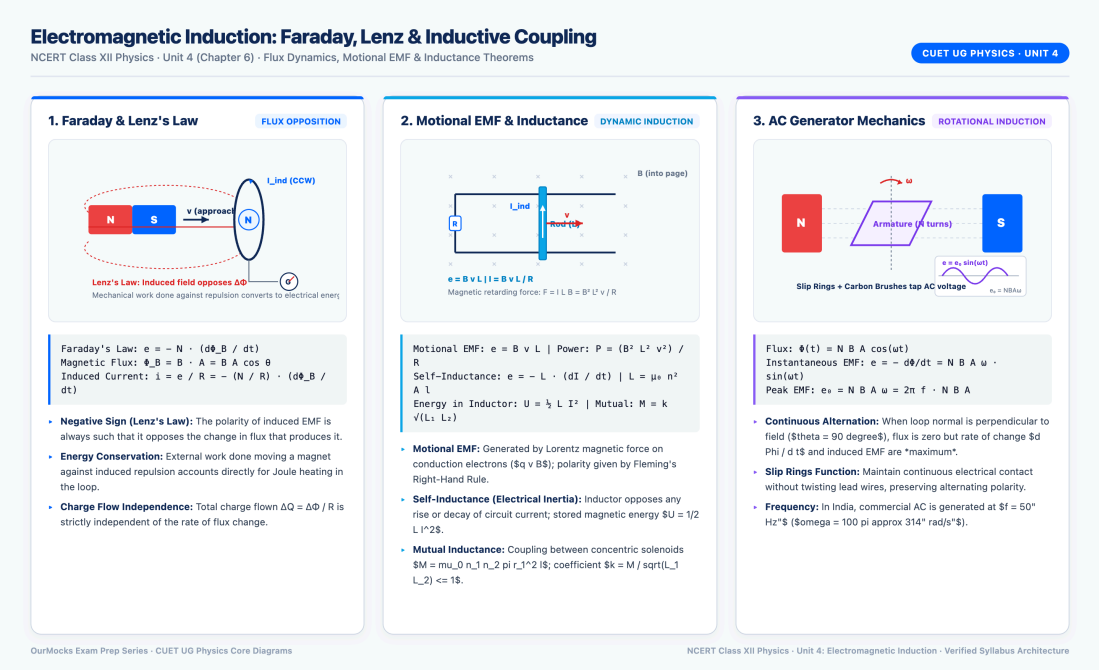


Figure 1: Electromagnetic Induction Dynamics: Faraday's flux change, Lenz's energy opposing reaction, and translational/rotational motional EMF.

Eddy Currents (Foucault Currents)

When bulk conductors are subjected to changing magnetic flux, induced circulating currents loop through the body of the conductor, termed **eddy currents**.

- **Phenomenon:** Eddy currents produce intense Joule heating (I^2R) and exert strong magnetic braking forces opposing motion.
- **Applications:**
  1. **Magnetic Braking in Trains:** Strong electromagnets induce eddy currents in rails/wheels, smoothly halting trains without mechanical wear.
  2. **Induction Furnace:** High-frequency alternating magnetic fields generate rapid eddy currents to melt metals cleanly in vacuum.
  3. **Dead-beat Galvanometer:** Coil wound on metallic copper/aluminium frame is brought to rest quickly without oscillatory swinging.
  4. **Electric Power Meters:** Rotating aluminium disc driven by eddy currents.
- **Minimization:** In transformers and induction motors, eddy current losses are minimized by replacing solid cores with **thin, laminated soft-iron sheets** coated with insulating varnish, slicing current loops into tiny paths.

WORKED EXAMPLE 2: MOTIONAL EMF ACROSS AIRCRAFT WINGS

Problem: A Boeing aircraft with a wingspan of 40 m flies horizontally northward at a speed of 900 km/h. If the Earth's total magnetic field at that location is 5.0×10^{-5} T and the angle of dip is 60°, find the potential difference developed between the wing tips.

Solution: Speed of aircraft:

$$v = 900 \times \left(\frac{5}{18}\right) = 250 \text{ m/s}$$

The aircraft flies horizontally. The vertical component of Earth's magnetic field (B_V) is cut by the horizontal wings:

$$B_V = B\sin\delta = (5.0 \times 10^{-5}) \times \sin 60° = 5.0 \times 10^{-5} \times \left(\frac{\sqrt{3}}{2}\right) = 4.33 \times 10^{-5} \text{ T}$$

Induced motional EMF across the wingspan ($l = 40$ m):

$$e = B_V v l = (4.33 \times 10^{-5}) \times 250 \times 40 = 4.33 \times 10^{-5} \times 10000 = 0.433 \text{ V}$$

The eastern wing-tip is at higher electrical potential by Fleming's Right-Hand Rule.

Self-Induction, Mutual Induction & Magnetic Energy

Self-Inductance (L)

Self-induction is the phenomenon in which an opposing EMF is induced in a single coil whenever the electric current passing through it varies with time. It is known as the **inertia of electricity**.

SELF-INDUCTANCE RELATIONS

Total magnetic flux linkage is proportional to current:

$$N\Phi = LI \Longrightarrow L = \frac{N\Phi}{I} \quad \text{[SI unit: Henry (H) or V} \cdot \text{s/A} = \Omega \cdot \text{s]}$$

By Faraday's law, induced back-EMF:

$$e = -L\frac{dI}{dt}$$

- 1 Henry is the self-inductance of a coil in which a rate of current change of 1 A/s induces an opposing back-EMF of 1 V.
- **Long Solenoid of length l, area A, and N turns (turn density $n = \frac{N}{l}$):**

$$B = \mu_0 nI \Longrightarrow \Phi = BA = \mu_0 nIA$$

$$L = \frac{N\Phi}{I} = \frac{N\left(\mu_0\left(\frac{N}{l}\right)IA\right)}{I} = \frac{\mu_0 N^2 A}{l} = \mu_0 n^2 Al$$

- If a core of relative permeability μ_r is inserted: $L = \mu_r\mu_0 n^2 Al$.

Mutual Inductance (M)

Mutual induction is the phenomenon where a time-varying current in one coil (primary) induces an EMF in an adjacent neighbouring coil (secondary):

MUTUAL INDUCTANCE RELATIONS

Flux linkage in coil 2 due to current in coil 1:

$$N_2\Phi_2 = M_{21}I_1 \Longrightarrow e_2 = -M_{21}\frac{dI_1}{dt}$$

By the Reciprocity Theorem: $M_{12} = M_{21} = M$.

- **Two Coaxial Long Solenoids (solenoid 1 inside solenoid 2, cross-section area A_1, length l):**

$$M = \mu_0 n_1 n_2 A_1 l = \frac{\mu_0 N_1 N_2 A_1}{l}$$

- **Coefficient of Coupling (k):**

$$M = k\sqrt{L_1 L_2} \quad (0 \le k \le 1)$$

$k = 1$ for perfect magnetic flux linkage (tightly coupled coils); $k = 0$ for perpendicular or widely separated coils.

Energy Stored in an Inductor & Energy Density

To establish a current I against the opposing self-induced back-EMF $e = -L\left(d\frac{i}{d}t\right)$, external work must be performed:

$$dW = |e|\ idt = \left(L\frac{di}{dt}\right)idt = Lidi$$

Integrating from $i = 0$ to $i = I$:

$$U = \int_0^I Lidi = \frac{1}{2}LI^2$$

- **Magnetic Energy Density (u_B):** For a solenoid, substituting $L = \mu_0 n^2 Al$ and $B = \mu_0 nI$:

$$U = \frac{1}{2}(\mu_0 n^2 Al)\left(\frac{B}{\mu_0 n}\right)^2 = \frac{B^2}{2\mu_0}(Al) \Longrightarrow u_B = \frac{U}{Al} = \frac{B^2}{2\mu_0} \quad [\text{J/m}^3]$$

WORKED EXAMPLE 3: SELF-INDUCTION AND STORED ENERGY IN A SOLENOID

Problem: An air-cored solenoid of length 0.50 m and cross-sectional area 20 cm^2 has 1000 closely wound turns. It carries a steady current of 4.0 A. Calculate: (a) the self-inductance of the solenoid, (b) the magnetic energy stored, and (c) the back-EMF if current drops to zero in 2.0 ms.

Solution: Given: $l = 0.50$ m, $A = 20 \times 10^{-4}$ m^2 $= 2.0 \times 10^{-3}$ m^2, $N = 1000$, $I = 4.0$ A. (a) Self-inductance:

$$L = \frac{\mu_0 N^2 A}{l} = \frac{(4\pi \times 10^{-7}) \times (1000)^2 \times (2.0 \times 10^{-3})}{0.50} = \frac{8\pi \times 10^{-4}}{0.50} = 1.6\pi \times 10^{-3}\ \text{H} \approx 5.03\ \text{mH}$$

(b) Stored magnetic energy:

$$U = \frac{1}{2}LI^2 = \frac{1}{2} \times (5.026 \times 10^{-3}) \times (4.0)^2 = 4.02 \times 10^{-2}\ \text{J} = 40.2\ \text{mJ}$$

(c) Back-EMF during cutoff:

$$|e| = L\frac{|\Delta I|}{\Delta t} = (5.026 \times 10^{-3}) \times \frac{4.0 - 0}{2.0 \times 10^{-3}} = 5.026 \times 2000 = 10.05 \text{ V}$$

Alternating Currents & AC Generator

AC Generator (Alternator)

An AC generator converts mechanical energy into alternating electrical energy based on Faraday's law of electromagnetic induction.

- **Construction:** Consists of an armature coil with N turns and area A rotating with constant angular velocity $\omega = 2\pi f$ in a uniform magnetic field (B) between two slip rings and carbon brushes.
- **Generated EMF:** Angle at time t is $\theta = \omega t$. Flux linkage: $\Phi(t) = NBA\cos(\omega t)$.

$$e(t) = -\frac{d\Phi}{dt} = -NBA(-\omega\sin(\omega t)) = NBA\omega\sin(\omega t)$$

$$e(t) = e_0\sin(\omega t) \quad [\text{Peak EMF: } e_0 = NBA\omega = 2\pi f NBA]$$

Mean and RMS Values of Alternating Voltage & Current

Alternating current varies sinusoidally: $I(t) = I_0\sin(\omega t)$, $V(t) = V_0\sin(\omega t)$.

Quantity	Mathematical Expression	Value Relative to Peak
Mean (Average) over Half Cycle	$I_{\text{avg}} = \frac{1}{\frac{T}{2}}\int_0^{\frac{T}{2}} I_0\sin(\omega t)dt$	$I_{\text{avg}} = \left(\frac{2}{\pi}\right)I_0 \approx 0.637 I_0$ (Zero over a full cycle T)
Root Mean Square (RMS / Effective)	$I_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T I_0^2\sin^2(\omega t)dt}$	$I_{\text{rms}} = \frac{I_0}{\sqrt{2}} \approx 0.707 I_0$ ($V_{\text{rms}} = \frac{V_0}{\sqrt{2}} \approx 0.707 V_0$)
Joule Heat Equivalence	$H = I_{\text{rms}}^2 Rt$	AC hot-wire instruments measure RMS values!

AC Applied to Pure Circuit Elements

1. Pure Resistor (R): $V = V_0\sin(\omega t)$ $I = \left(\frac{V_0}{R}\right)\sin(\omega t) = I_0\sin(\omega t)$

- Phase difference $\varphi = 0$.
- Current and voltage are in phase.
- Opposition: R (independent of ω).

2. Pure Inductor (L): $V = L\left(d\frac{I}{d}t\right) \Longrightarrow I = -\left(\frac{V_0}{\omega L}\right)\cos(\omega t)$ $I = I_0\sin\left(\omega t - \frac{\pi}{2}\right)$

- Voltage leads current by 90° ($\frac{\pi}{2}$).
- Inductive Reactance:

$$X_L = \omega L = 2\pi f L$$

- For DC ($f = 0$): $X_L = 0$ (short circuit).

3. Pure Capacitor (C): $q = CV \Longrightarrow I = d\frac{q}{d}t = \omega C V_0\cos(\omega t)$ $I = I_0\sin\left(\omega t + \frac{\pi}{2}\right)$

- Current leads voltage by 90° ($\frac{\pi}{2}$).
- Capacitive Reactance:

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

- For DC ($f = 0$): $X_C = \infty$ (blocks DC).

WORKED EXAMPLE 4: AC VOLTAGE AND REACTANCE CALCULATION

Problem: An AC source of $V = 220$ V, 50 Hz is connected across: (a) a pure inductor of $L = 0.20$ H, and (b) a pure capacitor of $C = 15\mu$F. Find the reactance and RMS current in each case.

Solution: Angular frequency: $\omega = 2\pi f = 2\pi \times 50 = 100\pi \approx 314.16$ rad/s. (a) For pure inductor:

$$X_L = \omega L = 314.16 \times 0.20 = 62.83\Omega$$

$$I_{\text{rms},L} = \frac{V_{\text{rms}}}{X_L} = \frac{220}{62.83} = 3.50 \text{ A}$$

(Current lags voltage by $\frac{\pi}{2}$ radians). (b) For pure capacitor:

$$X_C = \frac{1}{\omega C} = \frac{1}{314.16 \times 15 \times 10^{-6}} = \frac{10^6}{4712.4} = 212.2\Omega$$

$$I_{\text{rms},C} = \frac{V_{\text{rms}}}{X_C} = \frac{220}{212.2} = 1.037 \text{ A}$$

(Current leads voltage by $\frac{\pi}{2}$ radians).

Series LCR Circuit, Resonance & Quality Factor

Impedance and Phasor Analysis of Series LCR Circuit

In a series LCR circuit connected to an AC source $V = V_0 \sin(\omega t)$, the same instantaneous current $I = I_0 \sin(\omega t - \varphi)$ flows through all three elements.

- Phasor relationship: $(V) = (V)_R + (V)_L + (V)_C$.
- Since $(V)_L$ and $(V)_C$ are collinear and 180° out of phase, their net reactive phasor is $(V)_L - (V)_C$.
- Resultant voltage: $V_0^2 = V_R^2 + (V_L - V_C)^2 = (I_0 R)^2 + (I_0 X_L - I_0 X_C)^2 = I_0^2\left[R^2 + (X_L - X_C)^2\right]$.

IMPEDANCE AND PHASE ANGLE

Total Impedance (Z):

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \quad \text{[SI Unit: Ohm } (\Omega)]$$

Phase Angle (φ):

$$\tan\varphi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R} = \frac{\omega L - \frac{1}{\omega C}}{R}$$

- If $X_L > X_C$: Circuit is inductive ($\varphi > 0$), voltage leads current.
- If $X_C > X_L$: Circuit is capacitive ($\varphi < 0$), current leads voltage.
- If $X_L = X_C$: Circuit is purely resistive ($\varphi = 0$).

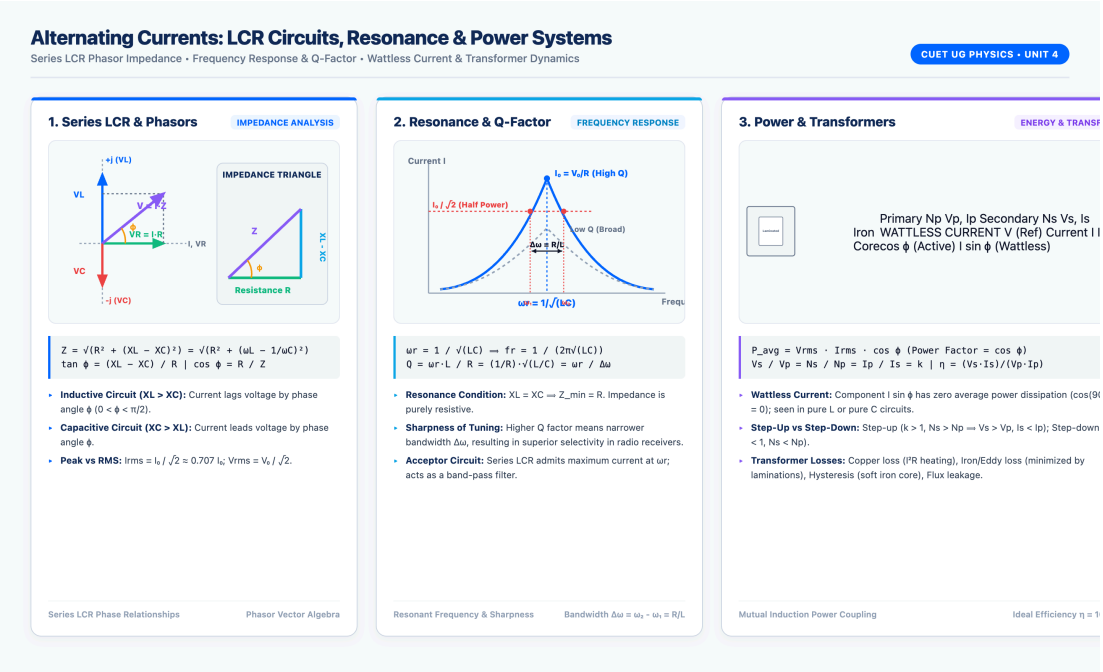


Figure 2: Alternating Current Analysis: Series LCR phasor triangle, frequency resonance response curve, and wattless power decomposition in transformers.

Electrical Resonance in Series LCR Circuit

Resonance occurs when the inductive and capacitive reactances cancel each other out ($X_L = X_C$):

SERIES RESONANCE FORMULAE

$$\omega_r L = \frac{1}{\omega_r C} \implies \omega_r^2 = \frac{1}{LC} \implies \omega_r = \frac{1}{\sqrt{LC}} \text{ rad/s}$$

$$f_r = \frac{\omega_r}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$

Characteristics of Series Resonance:

1. **Impedance is minimum:** $Z_{\min} = R$ (purely resistive).
2. **Current is maximum:** $I_0 = \frac{V_0}{R}$ (limited only by resistance R).
3. **Phase difference is zero:** $\varphi = 0$, voltage and current are in phase ($\cos\varphi = 1$).
4. **Acceptor Circuit:** Admits signals of resonant frequency while rejecting others; used in radio/TV tuning.

Quality Factor (Q-Factor) and Sharpness of Resonance

The sharpness of resonance determines how selectively the circuit discriminates against frequencies away from ω_r.

- **Half-Power Frequencies (ω_1, ω_2):** The frequencies at which circuit power drops to half its maximum value, where current amplitude falls to $\frac{I_0}{\sqrt{2}}$:

$$\Delta\omega = \omega_2 - \omega_1 = \frac{R}{L} \quad \text{[Bandwidth of the circuit]}$$

- **Definition of Q-Factor:**

$$Q = \frac{\text{Resonant Frequency}}{\text{Bandwidth}} = \frac{\omega_r}{\Delta\omega} = \frac{\frac{1}{\sqrt{LC}}}{\frac{R}{L}} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Also: $Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r CR} = \frac{V_L}{V} = \frac{V_C}{V}$ (Voltage magnification factor).

- Higher Q-factor indicates a sharper resonance curve, narrower bandwidth, and superior receiver selectivity.

WORKED EXAMPLE 5: SERIES LCR RESONANCE & Q-FACTOR

Problem: A series LCR circuit has $L = 2.0$ H, $C = 32\mu$F, and $R = 10\Omega$. Determine: (a) the resonant angular frequency ω_r, (b) the quality factor Q, and (c) the bandwidth $\Delta\omega$ of the circuit.

Solution: (a) Resonant angular frequency:

$$\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2.0 \times 32 \times 10^{-6}}} = \frac{1}{\sqrt{64 \times 10^{-6}}} = \frac{1}{8.0 \times 10^{-3}} = 125 \text{ rad/s}$$

(b) Quality factor Q:

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{10}\sqrt{\frac{2.0}{32 \times 10^{-6}}} = \frac{1}{10}\sqrt{\frac{1}{16 \times 10^{-6}}} = \frac{1}{10} \times \frac{1000}{4} = \frac{1}{10} \times 250 = 25$$

(c) Bandwidth $\Delta\omega$:

$$\Delta\omega = \frac{\omega_r}{Q} = \frac{125}{25} = 5.0 \text{ rad/s}$$

(Verification: $\Delta\omega = \frac{R}{L} = \frac{10}{2.0} = 5.0$ rad/s**).**

Power in AC Circuits & Transformers

Power in AC Circuits & Power Factor

Instantaneous power is $p(t) = V(t)I(t) = [V_0 \sin(\omega t)][I_0 \sin(\omega t - \varphi)]$. Integrating over a full time period T:

AVERAGE AC POWER & POWER FACTOR

$$P_{\text{avg}} = \frac{V_0 I_0}{2}\cos\varphi = \left(\frac{V_0}{\sqrt{2}}\right)\left(\frac{I_0}{\sqrt{2}}\right)\cos\varphi = V_{\text{rms}} I_{\text{rms}} \cos\varphi$$

- **Apparent Power:** $P_{\text{app}} = V_{\text{rms}} I_{\text{rms}}$ (measured in Volt-Amperes, VA).
- **True Power:** $P_{\text{true}} = V_{\text{rms}} I_{\text{rms}} \cos\varphi$ (measured in Watts, W).
- **Power Factor:** $\cos\varphi = \frac{\text{True Power}}{\text{Apparent Power}} = \frac{R}{Z}$.
- **Case 1 (Purely Resistive):** $\varphi = 0 \implies \cos\varphi = 1 \implies P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}}$ (maximum power dissipation).
- **Case 2 (Pure L or Pure C):** $\varphi = \pm\frac{\pi}{2} \implies \cos\varphi = 0 \implies P_{\text{avg}} = 0$ (zero net power dissipated over full cycle!).

Wattless Current

When AC flows through an ideal inductor or capacitor ($R = 0$, $\varphi = \frac{\pi}{2}$), no electrical power is dissipated despite current flow.

- The alternating current I_{rms} can be resolved into two orthogonal components relative to V_{rms}:
  1. **Active (Workful) Component:** $I_{\text{active}} = I_{\text{rms}} \cos\varphi$ in phase with voltage ($P = V_{\text{rms}} I_{\text{active}}$).
  2. **Reactive (Wattless) Component:** $I_{\text{wattless}} = I_{\text{rms}} \sin\varphi$ lagging/leading voltage by 90°.

$$P_{\text{wattless}} = V_{\text{rms}} (I_{\text{rms}} \sin\varphi) \cos(90°) = 0$$

- **Choke Coil:** An iron-cored inductor with high self-inductance L and low resistance R. It chokes (controls) AC current with negligible power loss ($\cos\varphi \approx 0$), far superior to power-wasting rheostats in fluorescent tubes.

The Transformer

A transformer is a static electromagnetic device that steps AC voltage up or down based on **mutual induction** between two insulated coils wound on a common laminated soft-iron core.

TRANSFORMER EQUATIONS

Let N_p and N_s be the number of turns in primary and secondary coils:

$$e_p = -N_p \frac{d\Phi}{dt}, \quad e_s = -N_s \frac{d\Phi}{dt}$$

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = k \quad \text{[Transformation Ratio]}$$

For an ideal (100% efficient) transformer: Input Power = Output Power ($V_p I_p = V_s I_s$):

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} = k$$

Efficiency (η):

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\% = \frac{V_s I_s \cos\varphi_s}{V_p I_p \cos\varphi_p} \times 100\%$$

Step-Up Transformer ($k > 1$):

- $N_s > N_p \implies V_s > V_p$.
- Current is stepped down: $I_s < I_p$.
- Secondary winding has many turns of thinner wire.
- Used at power generating stations to step voltage up to hundreds of kilovolts (132 kV – 400 kV) to minimize I^2R transmission line heating losses.

Step-Down Transformer ($k < 1$):

- $N_s < N_p \implies V_s < V_p$.
- Current is stepped up: $I_s > I_p$.
- Secondary winding has fewer turns of thicker wire.
- Used at local city substations and inside device adapters (converting 220 V AC down to 5 V – 12 V).

Energy Losses in Practical Transformers

1. **Copper Loss (I^2R Heating):** Joule dissipation in primary and secondary windings; minimized by using thick copper wire.
2. **Iron / Eddy Current Loss:** Circulating eddy currents in the magnetic core; minimized by using laminated sheets insulated with varnish.
3. **Hysteresis Loss:** Continuous cyclic magnetization and demagnetization of the core; minimized by using **soft iron** having narrow hysteresis loop.
4. **Flux Leakage:** Incomplete linkage between primary and secondary flux; minimized by winding primary and secondary coils concentrically over one another.

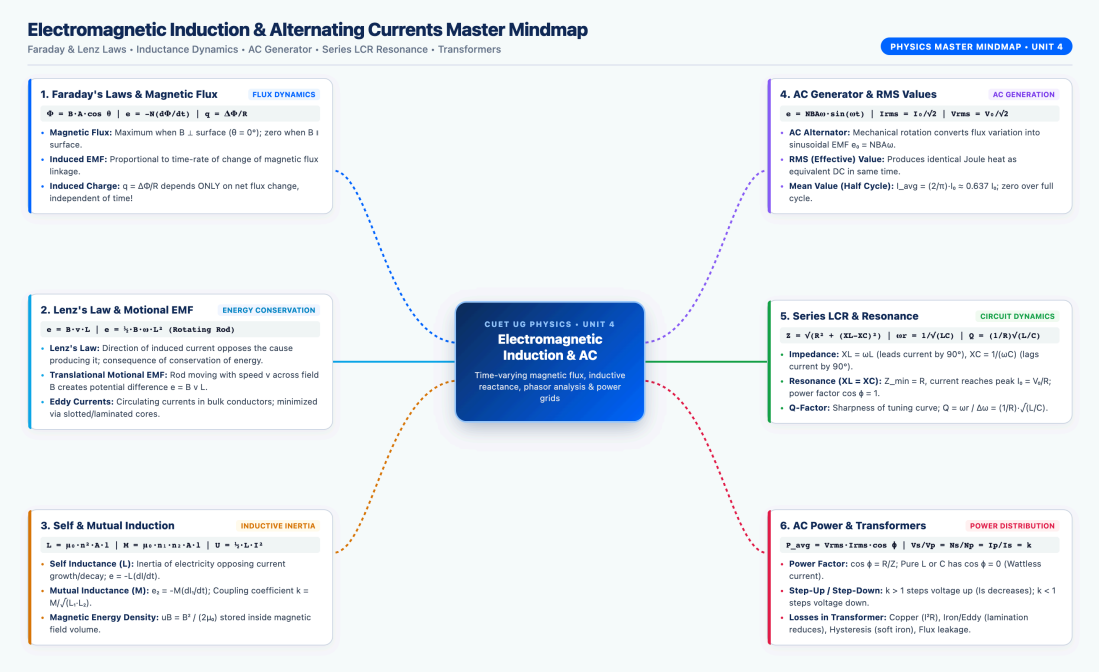


Figure 3: Master Synthesis Mindmap: Structural framework of Unit 4 Electromagnetic Induction and Alternating Currents.

MASTER EMI & ALTERNATING CURRENTS REVISION CHEATSHEET

- **Magnetic Flux & Faraday Law:** $\Phi = BA\cos\theta$ | $e = -N\left(d\frac{\Phi}{d}t\right)$ | Induced charge $q = \Delta\frac{\Phi}{R}$ (independent of time Δt).
- **Motional EMF:** Translating: $e = Bvl$ | Rotating rod about one end: $e = \frac{1}{2}B\omega l^2 = BfA$.
- **Inductances:** Solenoid $L = \mu_0 n^2 Al = \frac{\mu_0 N^2 A}{l}$ | Mutual $M = \mu_0 n_1 n_2 Al$ | Energy $U = \frac{1}{2}LI^2$ | $u_B = \frac{B^2}{2\mu_0}$.
- **AC RMS Relationships:** $I_{\text{rms}} = \frac{I_0}{\sqrt{2}} \approx 0.707 I_0$ | $V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$ | Half-cycle mean $I_{\text{avg}} = \left(\frac{2}{\pi}\right)I_0 \approx 0.637 I_0$.
- **Series LCR Impedance:** $Z = \sqrt{R^2 + (X_L - X_C)^2}$ | $X_L = \omega L$ | $X_C = \frac{1}{\omega C}$ | $\tan\varphi = \frac{X_L - X_C}{R}$.
- **Resonance Condition ($X_L = X_C$):** $\omega_r = \frac{1}{\sqrt{LC}}$ | $Z_{\min} = R$ | $I_0 = \frac{V_0}{R}$ | $Q = \left(\frac{1}{R}\right)\sqrt{\frac{L}{C}} = \frac{\omega_r}{\Delta\omega}$.
- **AC Power & Transformers:** $P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\varphi$ | Power factor $\cos\varphi = \frac{R}{Z}$ | $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} = k$.

CUET UG Exam-Style Practice Questions (15 MCQs)

Q1. A conducting circular loop of area 0.05 m^2 and resistance 2.0Ω is placed perpendicular to a magnetic field. If the magnetic field decreases from 0.60 T to 0.20 T in 0.10 s, the total electrical charge that flows through the loop is:

(A) 0.010 C (B) 0.020 C
(C) 0.10 C (D) 0.20 C

Sol. Q1. Total induced charge $\Delta q = \frac{\Delta\Phi}{R} = \frac{A\Delta B}{R}$. $\Delta B = 0.60 - 0.20 = 0.40$ T. $\Delta q = \frac{0.05\times0.40}{2.0} = \frac{0.020}{2.0} = 0.010$ C.

Answer: (A)

Q2. A metal rod of length 1.0 m rotates about one of its ends in a plane perpendicular to a uniform magnetic field $B = 0.50$ T at an angular velocity of 100 rad/s. The potential difference induced between the two ends of the rod is:

(A) 25 V (B) 50 V
(C) 100 V (D) 12.5 V

Sol. Q2. Induced EMF in a rotating rod: $e = \frac{1}{2}B\omega l^2 = \frac{1}{2} \times 0.50 \times 100 \times (1.0)^2 = 25$ V.

Answer: (A)

Q3. A long solenoid of radius R has n turns per unit length and carries a current $I = I_0 \sin(\omega t)$. A coil of N turns and radius r ($r < R$) is placed coaxially at its centre. The mutual inductance between the solenoid and the inner coil is:

(A) $\mu_0 nN\pi R^2$ (B) $\mu_0 nN\pi r^2$
(C) $\mu_0 n^2 N\pi r^2$ (D) $\frac{\mu_0 nN\pi r^2}{R}$

Sol. Q3. Field inside outer solenoid: $B = \mu_0 nI$. Flux linked with inner coil of radius r is $\Phi = B(\pi r^2) = \mu_0 nI\pi r^2$. Total flux linkage $N\Phi = \mu_0 nN\pi r^2 I \Longrightarrow M = \mu_0 nN\pi r^2$.

Answer: (B)

Q4. If the number of turns in an air-cored solenoid is doubled while its length and cross-sectional area remain unchanged, its self-inductance becomes:

(A) Doubled (B) Halved
(C) Four times (D) Unchanged

Sol. Q4. Self-inductance of solenoid $L = \frac{\mu_0 N^2 A}{l} \Longrightarrow L \propto N^2$. When N is doubled, $L' = \frac{(2N)^2}{N^2} \times L = 4L$.

Answer: (C)

Q5. Lenz's law of electromagnetic induction is a direct consequence of the law of conservation of:

(A) Charge (B) Linear Momentum
(C) Angular Momentum (D) Energy

Sol. Q5. Lenz's law ensures mechanical work done in moving a magnet against opposing electromagnetic forces is transformed into electrical/thermal energy, adhering to the Conservation of Energy.

Answer: (D)

Q6. In an AC circuit, an alternating voltage $V = 200\sqrt{2}\sin(100t)$ V is connected to a capacitor of capacitance 1.0μF. The RMS current in the circuit is:

(A) 10 mA **(B)** 20 mA
(C) $20\sqrt{2}$ mA **(D)** 40 mA

Sol. Q6. $V_{\text{rms}} = \frac{V_0}{\sqrt{2}} = \frac{200\sqrt{2}}{\sqrt{2}} = 200$ V. Angular frequency $\omega = 100$ rad/s. Capacitive reactance $X_C = \frac{1}{\omega C} = \frac{1}{100\times10^{-6}} = 10^4\Omega$. $I_{\text{rms}} = \frac{V_{\text{rms}}}{X_C} = \frac{200}{10^4} = 0.020$ A $= 20$ mA.

Answer: (B)

Q7. A series LCR circuit has $R = 8.0\Omega$, $X_L = 12.0\Omega$, and $X_C = 6.0\Omega$. The power factor of the circuit is:

(A) 0.60 **(B)** 0.50
(C) 1.0 **(D)** 0.80

Sol. Q7. Net reactance $X = X_L - X_C = 12.0 - 6.0 = 6.0\Omega$. Impedance $Z = \sqrt{R^2 + X^2} = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = 10\Omega$. Power factor $\cos\varphi = \frac{R}{Z} = \frac{8.0}{10.0} = 0.80$.

Answer: (D)

Q8. In a series LCR circuit at resonance, the phase difference between applied voltage and resulting current is:

(A) $\frac{\pi}{2}$ **(B)** π
(C) 0 **(D)** $\frac{\pi}{4}$

Sol. Q8. At resonance, $X_L = X_C$, so net reactance is zero and $Z = R$. Phase angle $\tan\varphi = \frac{X_L - X_C}{R} = 0 \Longrightarrow \varphi = 0$. Current and voltage oscillate in phase.

Answer: (C)

Q9. To double the resonant frequency of a series LCR circuit, the capacitance C should be changed to:

(A) $2C$ **(B)** $4C$
(C) $\frac{C}{2}$ **(D)** $\frac{C}{4}$

Sol. Q9. Resonant frequency $f_r = \frac{1}{2\pi\sqrt{LC}} \Longrightarrow f_r \propto \frac{1}{\sqrt{C}}$. To double f_r ($f_{r'} = 2f_r$), we need $\sqrt{\frac{C}{C'}} = 2 \Longrightarrow \frac{C}{C'} = 4 \Longrightarrow C' = \frac{C}{4}$.

Answer: (D)

Q10. The Quality factor (Q) of a series resonant circuit with inductance L, capacitance C, and resistance R is expressed by:

(A) $R\sqrt{\frac{C}{L}}$ **(B)** $\left(\frac{1}{R}\right)\sqrt{\frac{L}{C}}$
(C) $\left(\frac{1}{R}\right)\sqrt{\frac{C}{L}}$ **(D)** $R\sqrt{\frac{L}{C}}$

Sol. Q10. $Q = \omega_r \frac{L}{R} = \left(\frac{1}{\sqrt{LC}}\right)\left(\frac{L}{R}\right) = \left(\frac{1}{R}\right)\sqrt{\frac{L}{C}}$.

Answer: (B)

Q11. In an AC circuit, the instantaneous voltage is $V = 100\sin(100t)$ V and current is $I = 100\sin\left(100t + \frac{\pi}{3}\right)$ mA. The average power consumed is:

(A) 10 W (B) 2.5 W
(C) 5.0 W (D) $\frac{2.5}{\sqrt{2}}$ W

Sol. Q11. $P_{\text{avg}} = \frac{1}{2}V_0 I_0 \cos\varphi = \frac{1}{2} \times 100 \times (100 \times 10^{-3}) \times \cos\left(\frac{\pi}{3}\right) = \frac{1}{2} \times 10 \times 0.5 = 2.5$ W.

Answer: (B)

Q12. A step-down transformer transforms 2200 V down to 220 V. If primary coil has 5000 turns and efficiency is 90%, output current for 8.0 A input current is:

(A) 72 A (B) 80 A
(C) 88 A (D) 0.8 A

Sol. Q12. $\eta = \frac{V_s I_s}{V_p I_p} \Longrightarrow 0.90 = \frac{220 \times I_s}{2200 \times 8.0} = \frac{220 I_s}{17600} \Longrightarrow I_s = \frac{0.90 \times 17600}{220} = 72$ A.

Answer: (A)

Q13. A choke coil is preferred over a rheostat in controlling AC current in fluorescent lamps because:

(A) It increases voltage (B) It has zero inductance
(C) It minimizes power loss as heat (D) It acts as a rectifier

Sol. Q13. A choke coil has large inductance L and minimal resistance R ($\cos\varphi \approx 0$), throttling current with negligible Joule heat loss compared to a rheostat.

Answer: (C)

Q14. An alternating current of frequency 50 Hz flows through a 50Ω resistor. The time interval in which current drops from peak value to RMS value is:

(A) 2.5×10^{-3} s (B) 1.25×10^{-3} s
(C) 5.0×10^{-3} s (D) 1.0×10^{-2} s

Sol. Q14. $I(t) = I_0\cos(\omega t)$. At peak $t_1 = 0$. At RMS, $\cos(\omega t_2) = \frac{1}{\sqrt{2}} \Longrightarrow \omega t_2 = \frac{\pi}{4}$. $t_2 = \frac{\pi}{4\omega} = \frac{\pi}{4 \times 2\pi \times 50} = \frac{1}{400} = 2.5 \times 10^{-3}$ s.

Answer: (A)

Q15. A transformer core is laminated using thin sheets coated with varnish primarily to minimize:

(A) Hysteresis loss (B) Copper loss
(C) Eddy current loss (D) Flux leakage

Sol. Q15. Laminations break wide conducting loops into high-resistance narrow strips, reducing circulating eddy currents and associated I^2R core heating.

Answer: (C)